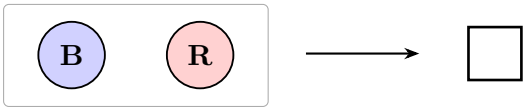
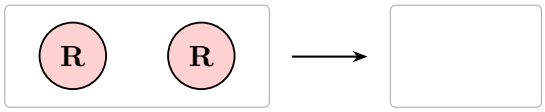
Use the same bag of three red counters and one blue. Draw twice in order, independently, giving every counter the same chance each time. Return and mix after each draw.

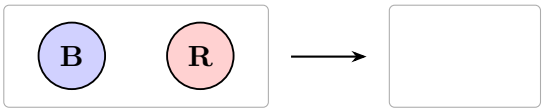


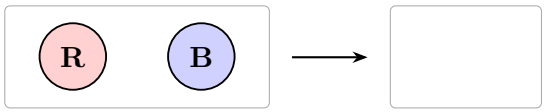
A sample rule: BR gives a square.

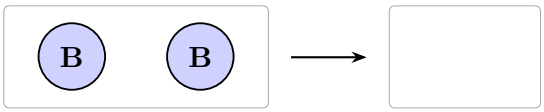
A rule gives each color pair a square, a circle, or no shape. A fair rule must sometimes give each shape, and give them equal chances when a shape comes out.

Problem 1: Invent a fair rule. Try it on six pairs, then decide whether your results are enough to settle that it is fair.

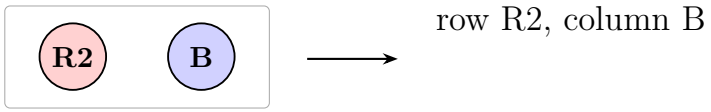






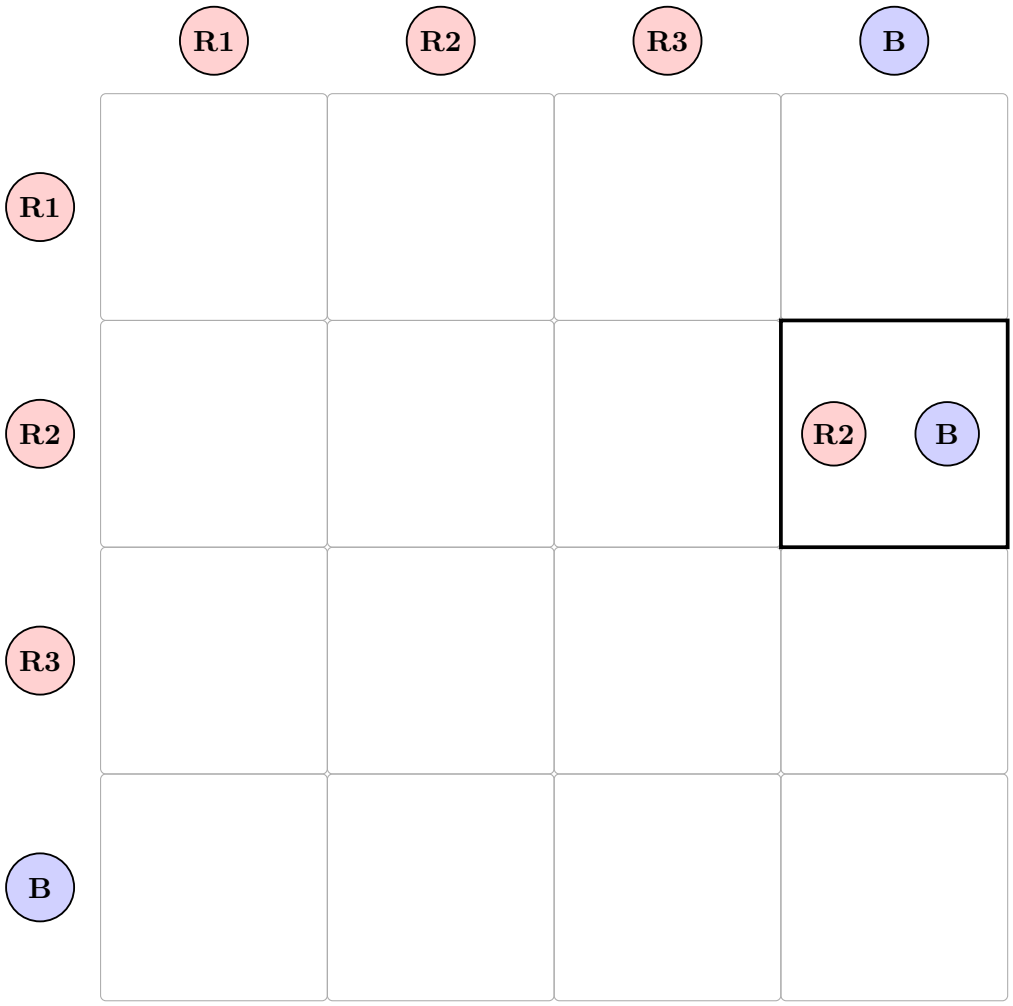


Problem 2: The counters are marked R1, R2, R3, and B. Use all the equally likely labeled pairs to check your rule exactly.



row R2, column B

First draw down the side; second draw across the top.



Each box names one equally likely pair of labeled counters.